Biochemistry Cume
December 6, 2002

Membrane Protein Structure (20 pts each, total 100pts)

I. A photosynthetic bacterium synthesizes a membrane protein (Mr 25,000 Da). X-ray analysis of the protein reveals that it consists of five parallel alpha-helical segments, each of which traverses the cytoplasmic membrane (thickness 6.0 nm).

[useful information: 1Å=10^{-1} nm, pitch of alpha helix= 1.5Å per residue, molecular weight per average amino acid = 110 Da]

(A.) Calculate the minimum number of amino acids necessary for one segment of the alpha-helix to completely traverse the membrane. Thus, what % of the protein must occur in the alpha-helical form?

(B.) The protein has been shown to be O-glycosylated. What are the likely amino acids to which the sugars would be attached?

(C.) The sugars are attached to an amino acid that is close (within 5 amino acids) to the N-terminus of the protein. What is likely organization of the protein in the membrane? A diagram showing the organization of the protein in the membrane would be an ideal answer.

II. A large number of G-protein coupled receptors (GPCRs; like rhodopsin) are encoded in the human genome. Their membrane-spanning regions consist of a seven alpha-helix bundle that is embedded in the lipid bilayer.

(A.) How could you predict the locations of the membrane spanning helices based on the sequences of GPCRs?

(B.) What are the forces/interactions that stabilize the folds of alpha-helical bundles within the lipid bilayer? How do these forces differ from those stabilizing the native folds of soluble proteins?